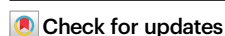


Fluorinated amorphous halides with improved ionic conduction and stability for all-solid-state sodium-ion batteries

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Designing halide solid electrolytes with high ionic conductivity and good (electro)chemical stability is essential for the advancement of all-solid-state sodium-ion batteries. Unfortunately, most sodium-based halide solid electrolytes experience limited ionic conductivities and ambiguous correlations between their structure features and ion transport properties. Here we report a design strategy to boost the conductivities of sodium halides by regulating vacancy and charge carrier concentrations through a facile Na- and Cl-deficient compositions method. This approach achieves a balanced structure with optimal vacancy and carrier content, rendering several-fold conductivities enhancement of series sodium halides. Furthermore, a fluorination-induced amorphization protocol is employed to enhance (electro)chemical stability and interfacial compatibility without detrimentally influencing conductivities. The promoted conductivities of the fluorinated sample are primarily due to increased local structural disorder and enhanced prismatic Na coordination. When paired with an uncoated $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ positive electrode and a Na_3PS_4 -coated $\text{Na}_{15}\text{Sn}_4$ negative electrode, the $\text{Na}_{0.5}\text{ZrCl}_4\text{F}_{0.5}$ catholyte enables the battery to run for 300 cycles, retaining 94.4% of its initial discharge capacity at room temperature. This study provides a versatile pathway for creating inorganic ion conductors with high conductivity and long-term cyclability, advancing the development of all-solid-state sodium-ion batteries.

All-solid-state sodium-ion batteries (ASSIBs) are poised to revolutionize grid-scale stationary storage due to their potential for higher energy density and enhanced safety over conventional liquid electrolyte systems^{1–3}. Moreover, the abundance and low cost of sodium raw materials further bolster the appeal of ASSIBs for large-scale applications^{4–6}. The key to their effective operation lies in the development of solid electrolytes (SEs) with high room-temperature (RT) ionic conductivity (σ), superior interfacial (electro)chemical stability with electrodes, and adequate deformability to ensure interface

affinity^{7–9}. Despite extensive research into various sodium SEs such as polymers, oxides, and sulfides, the practical deployment of ASSIBs has been hindered by the low RT conductivities of polymers, the high sintering temperatures required for oxides, and the narrow electrochemical windows and air sensitivity of sulfides^{10–12}. Recently, halide-based Li superionic conductors featuring monovalent anions (F^- , Cl^- , Br^- , I^-) have garnered attention for their high ionic conductivities ($>10^{-3} \text{ S cm}^{-1}$) at RT, high-voltage stability, and compressibility¹³. This has sparked interest in the development of halide-based Na-ion solid

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electrolytes (HNISEs) as potential alternatives. Nonetheless, the design principles of Li-ion halide SEs cannot be imprudently adopted for Na-ion analogs, mainly attributed to the much larger ionic radius of Na⁺ (1.06 Å) than Li⁺ (0.76 Å)¹⁴. Consequently, most reported HNISEs exhibit inadequate conductivities at RT ($\sim 10^{-5}$ S cm⁻¹). For instance, Na₃YCl₆ showed a conductivity of 9.85×10^{-8} S cm⁻¹ at RT, considerably lower than its Li-ion counterparts (Li₃YCl₆: 5.1×10^{-4} S cm⁻¹, Li₃YBr₆: 1.7×10^{-3} S cm⁻¹, RT) as reported by Asano in 2018^{15,16}. Additionally, Jung et al. proposed trigonal Na₂ZrCl₆ and orthorhombic NaAlCl₄ with moderate conductivities of 1.8×10^{-5} S cm⁻¹ and 3.9×10^{-6} S cm⁻¹ at 30 °C, respectively^{17,18}. The insufficient conductivities of HNISEs fundamentally challenge the capacity release and rate performance of ASSIBs.

In typical inorganic SEs, ion conduction primarily relies on vacancies and interstitial sites for ion hopping¹⁹. Increasing vacancy concentration in crystal lattices through hetero-valent doping has proven effective in enhancing conductivity by generating more sites for charge carrier transport³. For example, aliovalent Zr⁴⁺ substitution in Na_{2.25}Y_{0.25}Zr_{0.75}Cl₆¹⁵, Na_{2.4}Er_{0.4}Zr_{0.6}Cl₆²⁰, and Na_{2.25}Yb_{0.25}Zr_{0.75}Cl₆²¹ enhances RT conductivities to 6.6×10^{-5} S cm⁻¹, 3.5×10^{-5} S cm⁻¹, and 6.6×10^{-5} S cm⁻¹, respectively. Recently, Meng et al. designed a class of Na_{2.25-x}Y_{0.25}Zr_{0.75}Cl_{6-x} ($1.375 \leq x \leq 2$) SEs achieved with improved σ of 4.0×10^{-4} S cm⁻¹ (when $x = 1.625$) at RT, attributed to the presence of a mixed nanocrystalline and amorphous phase structure that promotes the formation of additional vacancies and free volumes²². Typically, creating vacancies in structures reduces carrier numbers, while higher carrier content often corresponds to fewer vacancies, indicating a strong negative correlation. However, current research tends to examine the effects of vacancy and carrier concentration separately, overlooking their potential interplay. The intricate relationship between charge carriers, vacancy content, and their combined impact on the bulk conductivity of HNISEs remains poorly understood.

In addition to the intrinsic conductivities of Na-ion SEs, their interface stability with electrodes is crucial for the efficient operation of ASSIBs. Incorporating fluorine into crystal structures is a promising modification tactic to improve this stability. Fluorine-doped ion conductors exhibit preferable electrochemical stability and interfacial compatibility due to the high electronegativity of F⁻, which facilitates the in-situ formation of an F-rich stable interphase between electrode and electrolyte interfaces²³⁻²⁵. Moreover, recent research has also shown that fluorinated chloride SEs exhibit moderate interface compatibility with sulfides²⁶. However, fluorinating inorganic ceramic SEs can reduce their conductivities owing to the robust ionic bonding between fluorine anions and mobile carriers, which hinders cation diffusion within the structures^{27,28}. Therefore, balancing ion transport and electrochemical stability in fluoride-doped HNISEs presents a significant challenge.

In this work, we precisely tune the vacancy concentration and Na-ion content within Na_{2+x}M_xZr_{1-x}Cl₆-type (M = Yb, Er, $x = 0, 0.25, 0.4$) HNISEs using a Na- and Cl-deficient compositions (abbreviated as Na/Cl-DC) strategy without adding any doping agents. Moreover, we investigate the effects of vacant-site and Na-ion contents on SEs' ion transport capability. Contrary to previous understandings^{3,12,29}, achieving a rational equilibrium between vacancy concentrations and mobile charge carrier numbers within HNISE structures is pivotal for ion migration efficiency. Building on this understanding, we develop a series of Na/Cl-DC HNISEs (Na_xZrCl_{4+x}, Na_xEr_{0.4}Zr_{0.6}Cl_{3.6+x}, and Na_xYb_{0.25}Zr_{0.75}Cl_{3.75+x}) that achieve the highest conductivities and lowest activation energy (E_a) under an optimized balance of vacant-sites and Na-ion. Additionally, we incorporate an appropriate proportion of F anions into the optimized Na_{0.5}ZrCl_{4.5-y}F_y structure. This modification enhances σ (Na_{0.5}ZrCl₄F_{0.5}, 1.1×10^{-4} S cm⁻¹ at 25 °C) through the fluorination-induced amorphization effect and provides good electrochemical compatibility with both electrodes and asymmetric sulfides SE interfaces. The enhanced Na⁺ mobility in fluorinated

amorphous SEs is mainly owing to increased local chemical structural disorder and enhanced prismatic Na coordination within the material. Using Na_{0.5}ZrCl₄F_{0.5} as the catholyte, coated-free Na₃V₂(PO₄)₃-based ASSIBs demonstrate stable cycling over 300 cycles at 0.1 C (11.7 mA g⁻¹) under RT. Our insights demonstrate favorable design flexibility and potential for developing high conductivity and (electro)chemically stable Na-ion SE systems.

Results and discussion

Ion transport scenarios under various vacancy and Na-ion concentration

The crystal structures of superionic conductors critically determine their ion transport capabilities and migration bottleneck sizes. In crystalline inorganic SEs, ionic hopping predominantly occurs between adjacent sites with the lowest energy barriers, whether through single-ion or multi-ion concerted migration³⁰. Furthermore, charge carriers typically conduct through the grain bulk and along boundaries via interconnected diffusion channels formed by the anion framework. Broadly, the conductivity of inorganic SEs can be succinctly expressed by Eq. (1):

$$\sigma = qn\mu \quad (1)$$

Where q , n , and μ are indexed to the elementary charge, charge carrier concentrations, and the mobility of carriers, respectively. Notably, carrier mobility is influenced by various factors, such as defect concentration, interatomic forces, disorder, and channel size³. In halogen anion frameworks, ion conduction mechanisms are shaped by vacancy and mobile charge carrier concentrations³¹, which typically exhibit a negative correlation. Excess charge carriers hinder ionic hopping, while too many vacancies increase migration distances. Thus, theoretical concentration-balanced thresholds for carriers and vacancies likely exist, with SEs achieving maximum conductivity when these thresholds are met. As illustrated in Fig. 1a–c, we disentangled the impact of carrier and vacant-site contents on Na-ion diffusion through three designed scenarios. In Fig. 1a, where Na-ion content > threshold (T_{Na^+}) and vacancy content < threshold (T_{vacancy}) in chloride frameworks, Na-ion hopping is hindered due to the limited number of available vacancies. Thus, the ion conduction process is dominated by vacancies. On the contrary, Fig. 1b shows that when Na-ion content < T_{Na^+} and vacancy content > T_{vacancy} , ion hopping is limited by the reduced mobility of Na-ion charge carriers. Consequently, the ion conduction process becomes ion-dominated. Figure 1c demonstrates that when both Na-ion and vacancy content are at their respective thresholds, Na-ion conduction is most efficient, indicating optimal ionic transport kinetics. Notably, a strong correlation exists between the concentration thresholds of vacancies and charge carriers in the structure. As vacancy content nears its threshold, charge carrier concentrations also approach their theoretical limits. Conversely, when these concentrations deviate from their thresholds, the variation trends of vacancies and charge carriers become diametrically opposite.

Meanwhile, the experimental results strongly support these viewpoints. As illustrated in Fig. 1d, optimizing vacancy and charge carrier contents using a Na/Cl-DC method yields compositions such as Na_{0.5}ZrCl_{4.5}, Na_{0.6}Er_{0.4}Zr_{0.6}Cl_{4.2}, and Na_{0.4}Yb_{0.25}Zr_{0.75}Cl_{4.15}, delivering higher conductivities of 8.1×10^{-5} S cm⁻¹, 5.7×10^{-5} S cm⁻¹, and 2.0×10^{-4} S cm⁻¹ than their original samples at 25 °C, respectively. As a proof-of-concept study, the representative trigonal phase structure (space group: P-3m1) Na_xZrCl_{4+x} SEs, noted for their resource abundance and cost-effectiveness relative to Er- and Y-based systems, were selected to research their structure changes induced by the Na/Cl-DC process. Then, the electrochemical behaviors of ASSIBs based above electrolyte systems will be systematically evaluated.

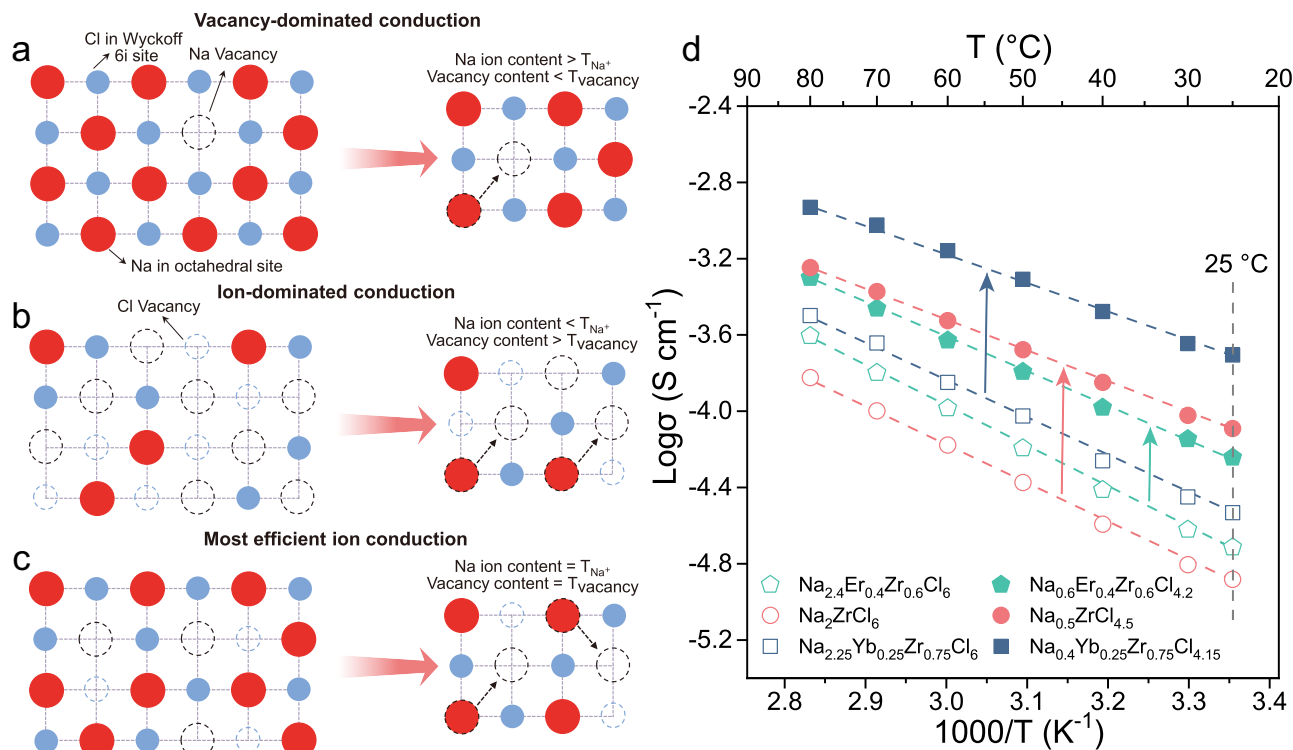


Fig. 1 | Schematic illustrates different effect scenarios of vacancy and Na-ion content on the ion conduction process. a With Na-ion content $> T_{Na^+}$ (threshold for Na^+ concentration) and vacancy content $< T_{vacancy}$ (threshold for vacancy concentration), the process is dominated by vacancies. **b** When Na-ion

content $< T_{Na^+}$ and vacancy content $> T_{vacancy}$, ion conduction predominates. **c** At Na-ion content $= T_{Na^+}$ and vacancy content $= T_{vacancy}$, respectively, conduction is the most efficient. **d** Arrhenius plots depict the ionic conductivities of different SE systems.

Structural analysis, micro-morphology, and ion conduction

Figure 2a shows the ternary Na-Zr-Cl composition diagram, highlighting the stoichiometric ratios explored in this study ($\text{Na}_x\text{ZrCl}_{4+x}$, where $0.2 \leq x \leq 2$) with red markers. The $\text{Na}_x\text{ZrCl}_{4+x}$ SEs were synthesized using a mechanochemical ball-milling method with varying molar ratios of NaCl and ZrCl_4 . The powder X-ray diffraction (XRD) patterns (Fig. 2b) revealed that the crystal structures of the $\text{Na}_x\text{ZrCl}_{4+x}$ series correspond to typical trigonal phases, isostructural with Li_2ZrCl_6 and Li_3YCl_6 ^{16,32}. Furthermore, as the NaCl molar ratio decreases, the XRD diffraction peaks of the samples show reduced intensity and a slight rightward shift. This suggests partial volume contraction in the crystal structure and lower product crystallinity. The observed volume shrinks in Na/Cl-DC samples is primarily attributed to the reduced presence of Na^+ (1.06 Å) and Cl^- (1.67 Å) ions within structures, which have larger ionic radii compared to the central Zr^{4+} (0.86 Å). Moreover, it can be found that the XRD peaks initially to the right and then back to the left as (x) changes from 2 to 0.2. This behavior is likely related to the dominant ion contributions during the Na/Cl-DC process. Specifically, when (x) decreases from 2 to 0.8, the lattice volume gradually shrinks (rightward shift) due to the reduction in the content of the dominant Na^+ and Cl^- ions. However, as (x) further decreases from 0.8 to 0.2, the lattice volume slightly expands (leftward shift), likely due to an increase in the relative content of the dominant Zr^{4+} ions within the structure. The weak crystallinity of the $\text{Na}_x\text{ZrCl}_{4+x}$ samples primarily resulting from reduced grain size and strain-broadening effects induced by the intense mechanochemical ball-milling. During the synthesis process, the electrolyte particles would be crushed and creating structural disorder and defects, which contributed to promoting ion hopping^{33,34}. The phase structure of the samples was further investigated using high-resolution transmission electron microscopy (HRTEM). As shown in Supplementary Fig. 1, the Na_2ZrCl_6 and $\text{Na}_{0.5}\text{ZrCl}_{4.5}$ electrolytes exhibit a combination of crystalline and

amorphous hybrid phase structures. Furthermore, the inductively coupled plasma optical emission spectrometer (ICP-OES) results yield chemical formulas of $\text{Na}_{1.98}\text{ZrCl}_{5.98}$ and $\text{Na}_{0.51}\text{ZrCl}_{4.51}$, closely aligning with the expected ingredient proportions (Supplementary Table 1), suggesting that no impurities or other composition were introduced during the ball milling process. To better elucidate the structural details of the crystalline phase, we performed Rietveld refinement based on the detectable XRD peaks of the samples, using the similar trigonal phase Li_2ZrCl_6 as the initial model to achieve maximum accuracy³². Figure 2c, d and Supplementary Figs. 2-3 illustrate the XRD Rietveld refinements and corresponding crystal structures of representative $\text{Na}_{0.5}\text{ZrCl}_{4.5}$ and Na_2ZrCl_6 SEs. Detailed structural parameters are provided in Supplementary Tables 2-3. The crystallographic atom sites within Na_2ZrCl_6 and $\text{Na}_{0.5}\text{ZrCl}_{4.5}$ SEs are identical, and both compounds share the same space group, $P\bar{3}m1$. This similarity indicates that the reduction in NaCl composition does not alter the electrolyte's microstructure, facilitating the examination of how vacant-site/Na-ion ratios affect the conductivities of HNISEs. Furthermore, Raman spectroscopy (Supplementary Fig. 4) reveals similar spectral profiles for Na_2ZrCl_6 and $\text{Na}_{0.5}\text{ZrCl}_{4.5}$ SEs, further supporting that the Na/Cl-DC process has minimal impact on the electrolyte structures. Distinctly, $\text{Na}_{0.5}\text{ZrCl}_{4.5}$ SEs exhibit more vacancies (Na and Cl vacant sites) and smaller lattice sizes compared to Na_2ZrCl_6 due to the Na/Cl-DC effect. The micro-morphologies of Na_2ZrCl_6 , $\text{Na}_{0.5}\text{ZrCl}_{4.5}$, and Na_3PS_4 SEs are compared by scanning electron microscopy (SEM) images (Supplementary Fig. 5). The $\text{Na}_{0.5}\text{ZrCl}_{4.5}$ products present smaller particle sizes and lower porosity (under 300 MPa) than the others, indicating reduced crystallinity and improved deformability in Na/Cl-DC SEs. Based on previous pertinent studies, a correlation has been established between electrolyte crystallinity and deformability, where a higher amorphous content within the structure typically enhances their deformability^{4,35,36}. One key factor influencing the

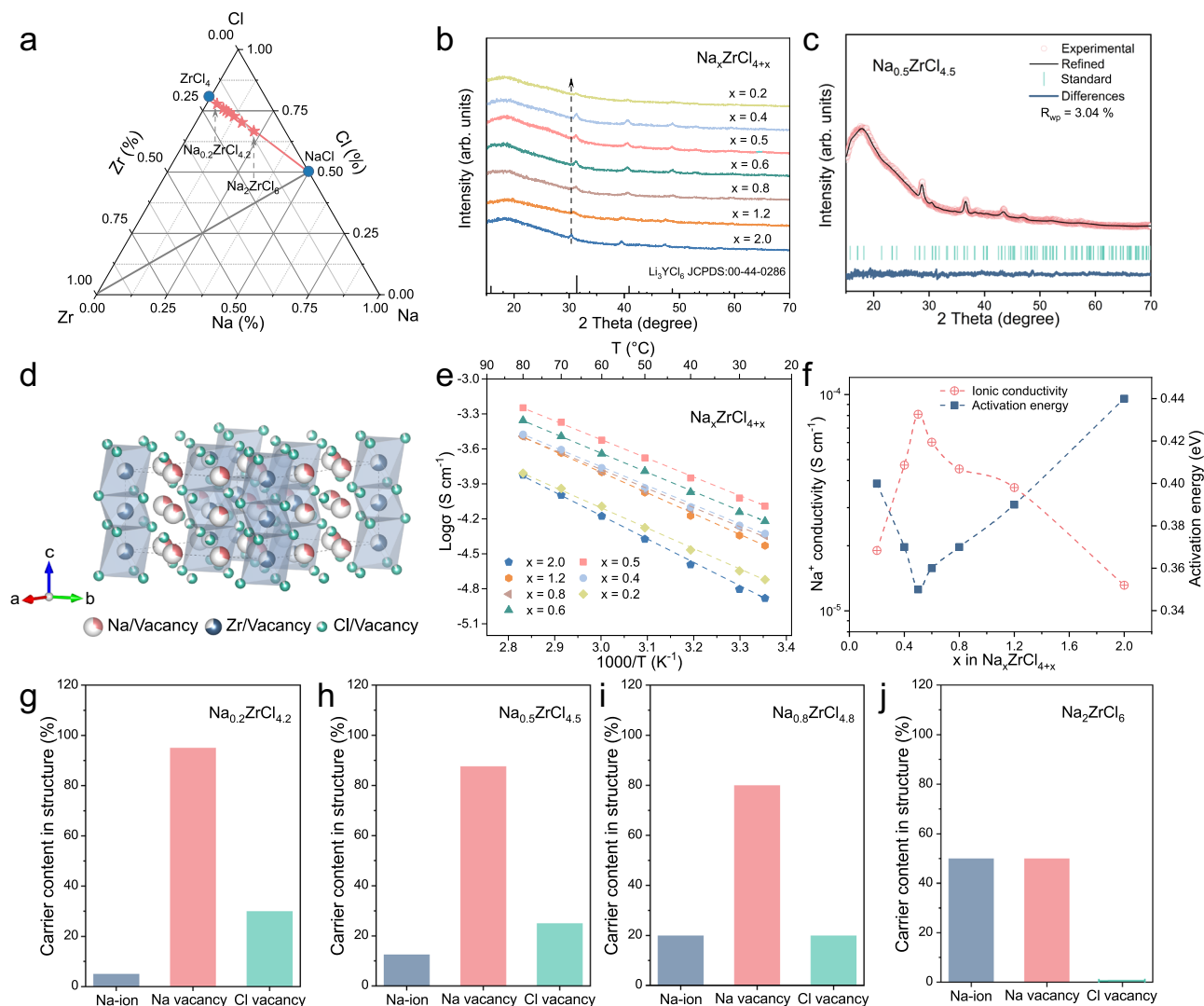


Fig. 2 | Structure and Na-ion transport characteristics of $\text{Na}_x\text{ZrCl}_{4+x}$ SEs. **a** The ternary composition diagram for Na-Zr-Cl illustrates the molar ratios adjusted in this study, highlighted in red. **b** XRD patterns for $\text{Na}_x\text{ZrCl}_{4+x}$ SEs, where $0.2 \leq x \leq 2$. **c** XRD Rietveld refinement results for $\text{Na}_{0.5}\text{ZrCl}_{4.5}$. **d** Crystal structure of $\text{Na}_{0.5}\text{ZrCl}_{4.5}$ derived from Rietveld refinement. **e** Arrhenius plots of various $\text{Na}_x\text{ZrCl}_{4+x}$ SE

systems. **f** Conductivity (σ) and activation energy (E_a) for $\text{Na}_x\text{ZrCl}_{4+x}$ systems at 25 °C. **g–j** Correlation between Na-ion content and vacancy concentration in $\text{Na}_x\text{ZrCl}_{4+x}$ SE systems across four compositions: **(g)** $\text{Na}_{0.2}\text{ZrCl}_{4.2}$, **(h)** $\text{Na}_{0.5}\text{ZrCl}_{4.5}$, **(i)** $\text{Na}_{0.8}\text{ZrCl}_{4.8}$, and **(j)** Na_2ZrCl_6 .

deformability of low-crystallinity materials may be the reduction of grain boundary effects⁴, which can mitigate microscopic physical contact loss within the electrolyte and increase relative density under applied pressure. However, more detailed investigations into grain boundaries and mechanical properties are beyond the scope of this study and will be explored in future research.

Figure 2e compares the Arrhenius plots of the $\text{Na}_x\text{ZrCl}_{4+x}$ ($0.2 \leq x \leq 2$) series based on electrochemical impedance spectra (EIS) at different measuring temperatures. In addition, the correlative σ and E_a , along with their EIS Nyquist plots of all products at 25 °C are demonstrated in Fig. 2f, Supplementary Fig. 6, and Supplementary Table 4. As a result, all Na/Cl-DC samples exhibit improved conductivities and reduced E_a . The optimized $\text{Na}_{0.5}\text{ZrCl}_{4.5}$ composition achieves the highest σ of $8.1 \times 10^{-5} \text{ S cm}^{-1}$ and lowest E_a of 0.35 eV at 25 °C, representing a more than six-fold enhancement compared to the pristine Na_2ZrCl_6 sample ($1.3 \times 10^{-5} \text{ S cm}^{-1}$ at 25 °C). Moreover, the electron conductivities of $\text{Na}_x\text{ZrCl}_{4+x}$ SEs are estimated to be approximately $10^{-9} \text{ S cm}^{-1}$ (Supplementary Fig. 7, Supplementary Table 8), indicating their favorable insulating properties. The improvement in conductivity of the Na/Cl-DC SEs likely arises from an increased number of vacant sites

within the structures, which provide more free volume and larger ion transport channels with reduced bottleneck size. To be noted, the conductivity change trend of $\text{Na}_x\text{ZrCl}_{4+x}$ SEs displays a distinct volcano-shaped curve, suggesting that neither insufficient nor excessive Na/Cl-DC within the structures (correlated with vacancy and carrier concentrations) is optimal for Na-ion diffusion.

To verify the aforementioned conjecture, we compared the changes in Na-ion content and vacant-site concentrations as mole number x increased (Fig. 2g–j, S8). The Na/Cl-DC condition in Na_2ZrCl_6 reduces the Na-ion content in the Cl sublattices while increasing the vacancy concentrations (both Na and Cl vacant sites). For example, when $x = 0.2$ ($\text{Na}_x\text{ZrCl}_{4+x}$), Na-ion in the $\text{Na}_{0.2}\text{ZrCl}_{4.2}$ structure occupies only 5% of the total ion sites, while the concentrations of Na and Cl vacant sites reach 95% and 30%, respectively (Fig. 2g). When x increases to 0.5, the electrolytes achieve their highest conductivity. At this point, both the Na-ion (12.5%) and vacancy contents (Na vacancy: 87.5%, Cl vacancy: 25%) reach their optimal thresholds (Fig. 2h). In stark contrast, further increasing the Na/Cl ion content ($x = 0.8, 2$) leads to decreased Na and Cl vacancy concentrations (Fig. 2i, j), resulting in a corresponding reduction in conductivity (Fig. 2e, f). Based on the

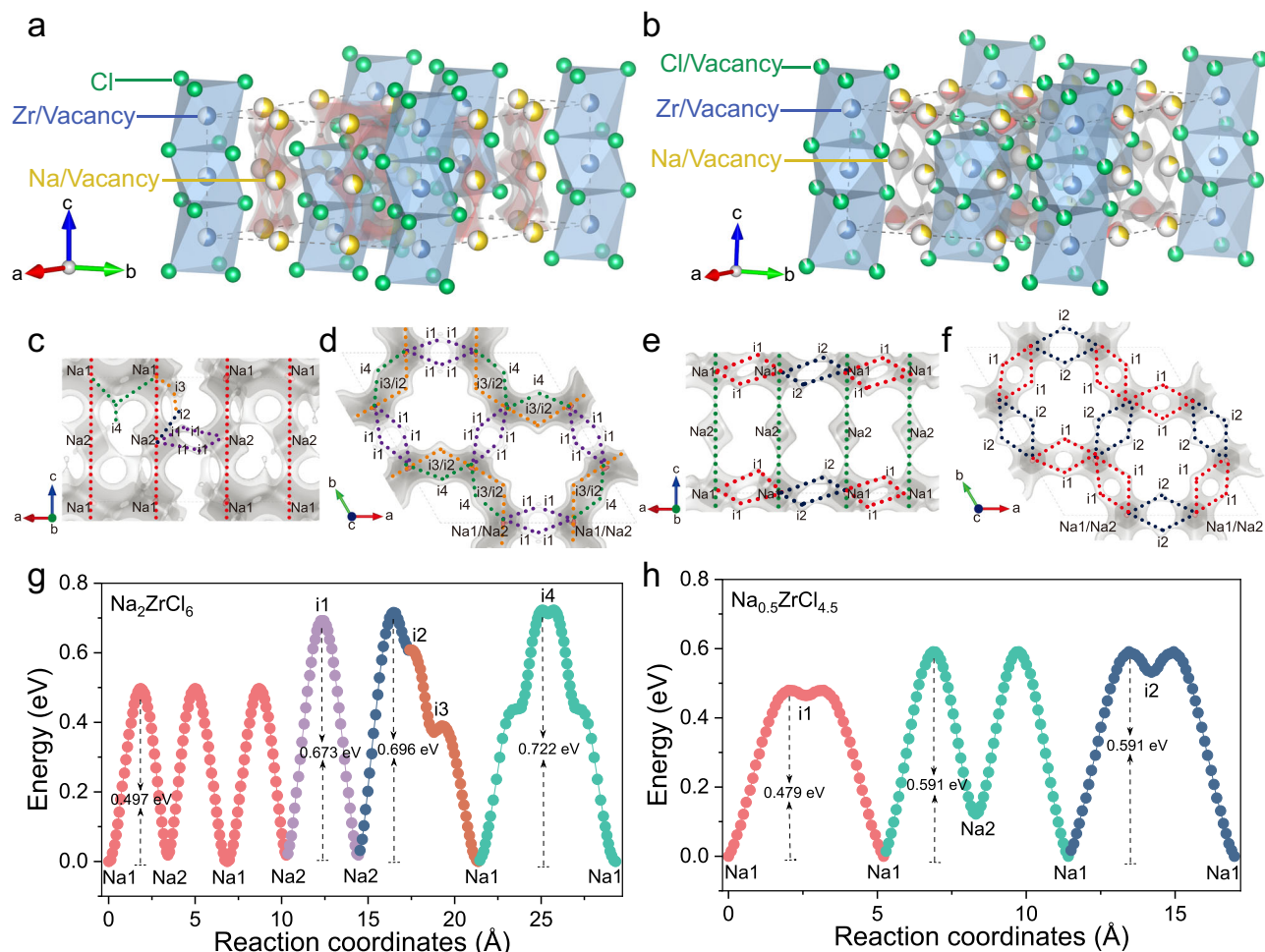


Fig. 3 | BVSE analysis of Na-ion conduction within the Na_2ZrCl_6 and $\text{Na}_{0.5}\text{ZrCl}_{4.5}$ structures. The crystal structures involved with Na-ion potential of (a) Na_2ZrCl_6 , and (b) $\text{Na}_{0.5}\text{ZrCl}_{4.5}$. The Na-ion conduction pathways within the structures of Na_2ZrCl_6 and $\text{Na}_{0.5}\text{ZrCl}_{4.5}$ are illustrated in panels (c, d) and (e, f), respectively. Different colored dashed lines represent distinct ion transport routes: Red dash:

Na1 – Na2 – Na1; Green dash: Na1 – i4 – Na1; Blue dash: Na2 – i2; Purple dash: Na2 – i1 – Na2; Orange dash: i2 – i3 – Na1 for Na_2ZrCl_6 , and Red dash: Na1 – i1 – Na1; Green dash: Na1 – Na2 – Na1; Blue dash: Na1 – i2 – Na1 for $\text{Na}_{0.5}\text{ZrCl}_{4.5}$. **g, h** Energy profiles corresponding to the ionic transport pathways are shown in panels (c–f), with color coding to match the pathways.

above analysis, we concluded that an optimal balance between mobile charge carrier and vacancy concentrations is crucial for efficient ion conduction in sodium superionic conductors. Building on this inference, we developed a series of Na-ion halide SEs systems with improved conductivities by adjusting carrier and vacancy contents through facile Na/Cl-DC strategies. As demonstrated in Supplementary Figs. 9–10 and Supplementary Tables 5–6, the optimized compositions of $\text{Na}_{0.6}\text{Er}_{0.4}\text{Zr}_{0.6}\text{Cl}_{4.2}$ and $\text{Na}_{0.4}\text{Yb}_{0.25}\text{Zr}_{0.75}\text{Cl}_{4.15}$ exhibited decent conductivities of $5.7 \times 10^{-5} \text{ S cm}^{-1}$ and $2.0 \times 10^{-4} \text{ S cm}^{-1}$, respectively, compared to their pristine counterparts ($\text{Na}_{2.4}\text{Er}_{0.4}\text{Zr}_{0.6}\text{Cl}_6$: $1.9 \times 10^{-5} \text{ S cm}^{-1}$; $\text{Na}_{2.25}\text{Yb}_{0.25}\text{Zr}_{0.75}\text{Cl}_6$: $2.9 \times 10^{-5} \text{ S cm}^{-1}$) at 25°C . The above results clearly demonstrate that enhancing the conductivity of inorganic SEs by balancing vacancy and carrier concentrations is a universal and feasible strategy.

To elucidate the effect of electrolyte structural features on ion conduction, the bond valence site energy (BVSE) method was employed to calculate ionic transport channels and corresponding migration energy barriers. In the Na_2ZrCl_6 structure (Fig. 3a), the [Na1–Na2–Na1] chain along the [001] direction was identified as the most efficient 1D transport channel, with a migration barrier of 0.497 eV (marked in red in Fig. 3c–d and g). Additionally, the Na-ion can also migrate along the [Na2–i1–Na2], [Na2–i2–i3–Na1], and [Na1–i4–Na1] directions (marked in purple, blue/orange, and green, respectively, in Fig. 3c, d, g). These paths connect the [Na1–Na2–Na1] chain to form a 3D

transport network, albeit with higher migration barriers of 0.673 eV, 0.696 eV, and 0.722 eV, respectively. In contrast, the $\text{Na}_{0.5}\text{ZrCl}_{4.5}$ structure (Fig. 3b) exhibits the lowest ion migration barrier of 0.479 eV along the [Na1–i1–Na1] direction (marked in red in Fig. 3e, f, h), effectively interconnecting the 1D [Na1–Na2–Na1] transport channels to form the most efficient 3D ionic percolating networks. Moreover, the other ionic transport channels along the [Na1–Na2–Na1] and [Na1–i2–Na1] directions (marked in green and blue, respectively, in Fig. 3e, f, h) also exhibit relatively low migration barriers of 0.591 eV. These findings further corroborate the more favorable Na-ion transport environments within Na/Cl-DC phase structures. It should be noted that BVSE analysis primarily characterizes ionic diffusion behaviors in crystal structures with periodic atomic arrangements and does not account for contributions from other non-ideal factors, such as amorphous matrices, defects, and surface structures. These features are typically nonperiodic owing to the robust ball-milling reactions. However, the crystalline component in Na/Cl-DC samples delivers lower migration energy barriers compared to pristine samples prepared under the same conditions. Thus, the effective transport barriers calculated from BVSE analysis provide powerful evidence of the improved conductivities in Na/Cl-DC SEs. Furthermore, nonperiodic compositions likely play an important role in ionic migration, warranting systematic research applying advanced characterization techniques with exceptionally fine spatial resolution, such as aberration-corrected

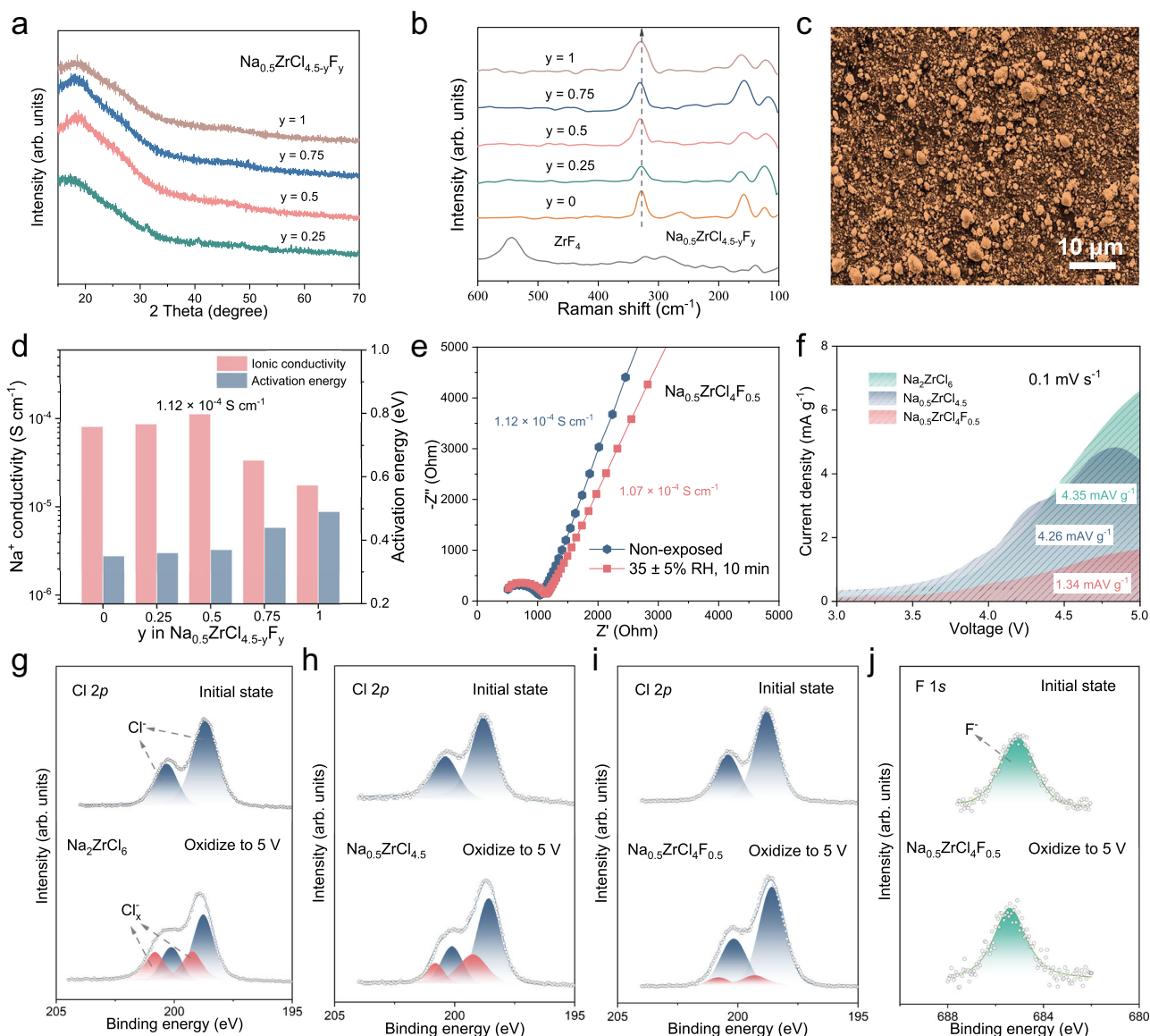


Fig. 4 | HNISEs for F-substituted $\text{Na}_{0.5}\text{ZrCl}_{4.5-y}\text{F}_y$ ($\text{Na}_{0.5}\text{ZrCl}_4\text{F}_{0.5}$). **a** XRD patterns of $\text{Na}_{0.5}\text{ZrCl}_{4.5-y}\text{F}_y$ ($0 < y \leq 1$) SEs. **b** Raman spectrum for $\text{Na}_{0.5}\text{ZrCl}_{4.5-y}\text{F}_y$ and ZrF_4 powders. To more clearly compare Raman peak shifts, we normalized the y-axis intensity. **c** SEM images of fluorinated particle samples. False coloring was utilized on the images to distinctly highlight the micro-morphology of the particles. **d** The σ

and corresponding E_a for $\text{Na}_{0.5}\text{ZrCl}_{4.5-y}\text{F}_y$ SEs at 25 °C. **e** EIS Nyquist plots for $\text{Na}_{0.5}\text{ZrCl}_4\text{F}_{0.5}$ before and after exposure to ambient humidity (35 ± 5% RH) for 10 min. **f** CV curves for asymmetric $\text{Na}_{15}\text{Sn}_4$ | NPS | HNISEs | HNISEs-VGCF cells at a scan rate of 0.1 mV s⁻¹. **g–j** XPS results for halide SEs Na_2ZrCl_6 (**g**), $\text{Na}_{0.5}\text{ZrCl}_{4.5}$ (**h**), and $\text{Na}_{0.5}\text{ZrCl}_4\text{F}_{0.5}$ (**i**, **j**) before and after oxidation to 5 V.

transmission electron microscopy³². Additionally, the conductivities of Na/Cl-DC electrolyte systems could be further improved by optimizing the proportions of nonperiodic components within the structures. For typical glass-ceramic electrolytes, such as Na halides studied in this work, ion conduction is predominantly influenced by the bulk phase, amorphous phase, and the interfaces or grain boundaries within the microstructure^{14,26}. However, quantifying the specific contributions of these different components to ion conduction presents significant challenges with current technologies. We contend that investigating the synergistic ion conduction among the crystalline region, amorphous region, and the interface between them warrants further in-depth study in the future.

The general design strategy of the HNISEs in enhancing (electro) chemical stability

Ordinarily, the fluorine-ion substitution can effectively improve the (electro)chemical stability of HNISEs but at the cost of sacrificed

conductivities due to the lower polarizability and smaller ionic radii of F⁻ (119 pm) compared to Cl⁻ (167 pm) or Br⁻ (182 pm)³⁷. Interestingly, in this study, the reduction in conductivity of fluorinated HNISEs can be well mitigated by the amorphization-induced protocol. Therefore, to further improve the electrochemical compatibility of Na/Cl-DC SEs with electrodes, varying amounts of F⁻ were incorporated into the $\text{Na}_{0.5}\text{ZrCl}_{4.5-y}\text{F}_y$ ($0 \leq y \leq 1$) series through mechanochemical synthesis. XRD results (Fig. 4a) show that the pattern peaks diminish as the F⁻ content increases, indicating that these fluorinated SEs tend to become amorphous during the synthesis procedures. Additionally, the HRTEM of $\text{Na}_{0.5}\text{ZrCl}_4\text{F}_{0.5}$ samples clearly displays amorphous characteristics (Supplementary Fig. 1), evidenced by the absence of distinct lattice fringes, which aligns with the XRD results. The mixing of multiple anions can induce localized structure disorder and rearrangements within the lattice, which likely contributes to the formation of an amorphous matrix and the production of smaller nanosized crystallites^{35,38}. Concurrently, Raman spectrums further suggest the

successful fluorination of $\text{Na}_{0.5}\text{ZrCl}_{4.5}$ without introducing additional impurities (Fig. 4b). Furthermore, the peak positions in the spectra of fluorinated samples emerge slight left shift (higher wavenumber) compared to pristine $\text{Na}_{0.5}\text{ZrCl}_{4.5}$ (~ 328 , 158 , and 124 cm^{-1}), primarily due to the formation of shorter F–Zr bonds within the structures. The SEM image (Fig. 4c) of $\text{Na}_{0.5}\text{ZrCl}_4\text{F}_{0.5}$ reveals small particle sizes ($< 5\text{ }\mu\text{m}$) consisting of irregular micro- and nano-particles. The reduced average particle and non-crystalline structure result in denser cold-pressed pellets ($\sim 98.2\%$ relative density, Supplementary Table 9) with decreased porosity (Supplementary Fig. 11a). The improved compressibility of the powder samples is more favorable for intimate solid-solid interface contact between electrode materials and SEs³⁶. Furthermore, the corresponding energy-dispersive X-ray spectroscopy (EDS) mapping implies the uniform dispersion of Zr, Cl, and F elements within the samples, indicating successful chemical fluorination of $\text{Na}_{0.5}\text{ZrCl}_{4.5}$ rather than physical mixing of fluorides (Supplementary Fig. 11b–d).

The ion conduction properties of the $\text{Na}_{0.5}\text{ZrCl}_{4.5-y}\text{F}_y$ ($0 \leq y \leq 1$) compositional series were assessed via EIS and Arrhenius plots (Supplementary Fig. 12a–b). The conductivities of $\text{Na}_{0.5}\text{ZrCl}_{4.5-y}\text{F}_y$ SEs reveal a pattern characterized by an initial increase followed by a subsequent decrease. At an F doping content near 0.5, the electrolyte achieves the highest σ of $1.12 \times 10^{-4}\text{ S cm}^{-1}$ at 25°C (Fig. 4d). Furthermore, the $\text{Na}_{0.5}\text{ZrCl}_4\text{F}_{0.5}$ SEs display electron conductivities of approximately $10^{-10}\text{ S cm}^{-1}$ (Supplementary Fig. 12c, Supplementary Table 8), exhibiting decent electron insulating properties. In brief, when the F doping content (y) is ≤ 0.5 , the contribution of the amorphous effect to ion transport exceeds the constrained force between F^- anion and Na^+ cation. Thus, the conductivities of $\text{Na}_{0.5}\text{ZrCl}_{4.5-y}\text{F}_y$ ($0 < y \leq 0.5$) SEs exhibited a growth trend. Whereas, for $y > 0.5$, the conductivities dramatically decrease because of the stronger attraction between F^- and carriers. Recent studies have demonstrated that the amorphization of SEs can significantly improve their ion conduction capabilities, owing to the unique local chemistry environments and ion migration mechanisms in amorphous domains^{11,38–41}.

The local chemical structure of the samples was further characterized using ^{23}Na solid-state nuclear magnetic resonance (ss-NMR). As shown in Supplementary Fig. 13, in the $\text{Na}_{0.5}\text{ZrCl}_4\text{F}_{0.5}$ sample, ^{23}Na signals linked to octahedral Na environments broaden and decrease significantly in intensity, while those for prismatic Na signals dramatically increase compared to pristine $\text{Na}_{0.5}\text{ZrCl}_{4.5}$. This indicates a preference for prismatic sites upon fluorination. A broadened shoulder peak at $\sim 3.8\text{ ppm}$ suggests increased Zr^{4+} disorder in octahedral Na environments⁴². The more Na in prismatic environments and increased Zr site disorder in fluorinated electrolytes support rapid Na^+ migration via isotropic conduction pathways and enhanced Na/vacancy disorder^{22,43}. To elucidate the anion framework changes induced by fluorination-driven amorphization, X-ray absorption near-edge structure (XANES) spectroscopy was employed to characterize the local structure of HNISEs. Supplementary Fig. 14 presents Zr K-edge XANES spectra of HNISEs, analyzed to link average local chemical environments to Na-ion migration within the amorphous phase. The Zr K-edge main edge position ($\sim 18020\text{ eV}$) across all samples confirms the quadrivalent oxidation state of Zr^{4+} . The whiteness of Na_2ZrCl_6 and $\text{Na}_{0.5}\text{ZrCl}_{4.5}$ exhibits three distinct splits (Supplementary Fig. 14a), resembling the whiteness of ZrCl_4 , indicating a similar $(\text{ZrCl}_6)^{2-}$ octahedral coordination with unique Zr–Cl bond lengths ($> 0.16\text{ }\text{\AA}$ between each other). In contrast, the whiteness of $\text{Na}_{0.5}\text{ZrCl}_4\text{F}_{0.5}$ smooths and merges, reflecting more uniform Zr–Cl/F bond lengths and suggesting a disordered and distorted anion structure³⁹. Fourier-transformed (FT) extended X-ray absorption fine structure (EXAFS) spectroscopy further quantifies the coordination environment around Zr (Supplementary Fig. 14b). Peak intensity change from F and Cl scatterers around Zr reveals increased Zr–F coordination and reduced Zr–Cl coordination, confirming nearest Cl

substitution by F upon fluorination. Wavelet-transformed (WT) EXAFS analysis (Supplementary Fig. 14c) further corroborates this, showing Zr coordination with F, Cl, and Zr at -1.5 , 1.9 , and $3.3\text{ }\text{\AA}$, respectively, indicating a mixed anion polyhedron $[\text{ZrCl}_4\text{F}_2]^{(a+b-4)-}$. Conversely, $\text{Na}_{0.5}\text{ZrCl}_{4.5}$ exhibits only Zr–Cl and Zr–Zr signals, characteristic of $(\text{ZrCl}_6)^{2-}$ octahedra (Supplementary Fig. 14d). Compared to Na_2ZrCl_6 and $\text{Na}_{0.5}\text{ZrCl}_{4.5}$, $\text{Na}_{0.5}\text{ZrCl}_4\text{F}_{0.5}$ demonstrates a more disordered Zr–F/Cl polyhedron with averaged Zr–X ($X = \text{Cl}, \text{F}$) bond lengths, contributing to flattening the energy landscape for Na-ion migration.

Moreover, the change in Na–F bond lengths after fluorination-driven amorphization may potentially impact Na^+ mobility. Here we employed *ab initio* molecular dynamics (AIMD) simulations to investigate Na–F bond length variations in the amorphous fluorinated electrolyte (Supplementary Fig. 15). The radial distribution functions (RDF) and static crystallographic data reveal that increased amorphousness results in longer Na–F bonds, while higher crystallinity leads to shorter Na–F bonds. These elongated Na–F bonds weaken coulombic interactions between the anion framework and mobile Na^+ , thus facilitating rapid Na^+ migration. Consequently, the observed improvement in conductivity of samples with partial F^- substitution is fundamentally attributed to local chemical structural disorder, an increase in prismatic Na coordination, and changes in bond lengths within the amorphous fluorinated SEs. However, excessive F^- tends to form robust bonds with mobile charges, impeding the effective diffusion of Na^+ within crystal structures. Given the favorable balance between ion conduction and (electro)chemical stability in fluorinated SEs, the optimized $\text{Na}_{0.5}\text{ZrCl}_4\text{F}_{0.5}$ samples will be the primary focus of subsequent research.

Another critical challenge for HNISEs is their sensitivity to moisture due to the deliquescent properties of most halides⁴⁴. In this work, the moisture stability of various SEs was assessed using EIS Nyquist plots and XRD patterns. After 10 min of exposure to ambient relative humidity (RH, $35 \pm 5\%$), the XRD pattern of $\text{Na}_{0.5}\text{ZrCl}_4\text{F}_{0.5}$ showed no additional impurity peaks (Supplementary Fig. 16). Furthermore, the conductivity of $\text{Na}_{0.5}\text{ZrCl}_4\text{F}_{0.5}$ remained stable after humidity exposure (Fig. 4e), demonstrating good moisture tolerance. In contrast, the XRD pattern of the $\text{Na}_{0.5}\text{ZrCl}_{4.5}$ pellet delivered attenuated peak intensity and reduced conductivity following similar moisture treatments (Supplementary Fig. 17). The improved humidity stability of $\text{Na}_{0.5}\text{ZrCl}_4\text{F}_{0.5}$ can be ascribed to the stronger Zr/Na–F bonds formed within the structures through F substitution^{24,45,46}.

The electrochemical stability of $\text{Na}_{0.5}\text{ZrCl}_4\text{F}_{0.5}$ was evaluated using cyclic voltammetry (CV) tests in an asymmetric configuration. Vapor-grown carbon fibers (VGCF)/SEs composites were used as the working electrode, while a $\text{Na}_{15}\text{Sn}_4$ (0.1 V vs Na|Na^+) alloy served as the reference electrodes. In composite electrodes, the incorporation of VGCF allows for continuous and efficient electron transport, enabling precise monitoring of electrochemical reaction signals⁴⁷. Figure 4f demonstrates the positive scan curves for various SEs at a scan rate of 0.1 mV s^{-1} . Notable, $\text{Na}_{0.5}\text{ZrCl}_4\text{F}_{0.5}$ exhibits the lowest integrated current of 1.34 mAV g^{-1} , in contrast to Na_2ZrCl_6 (4.35 mAV g^{-1}) and $\text{Na}_{0.5}\text{ZrCl}_{4.5}$ (4.26 mAV g^{-1}), manifesting the fluorination process significantly enhances the oxidation stability of HNISEs within high-voltage regimes. X-ray photo-electron spectroscopy (XPS) was employed to assess the high-voltage stability of these HNISEs. When oxidation is up to 5 V , oxidation peaks related to Cl_x^- are evident in the spectra of Na_2ZrCl_6 and $\text{Na}_{0.5}\text{ZrCl}_{4.5}$, indicating their relative instability at elevated voltages (Fig. 4g, h). In contrast, the spectra for $\text{Na}_{0.5}\text{ZrCl}_4\text{F}_{0.5}$ (Fig. 4i, j) show no obvious oxidation peaks for Cl_x^- and F_x^- , highlighting its good anti-oxidation properties attributed to the stronger electronegativity of fluorine. Although current technologies do not permit precisely determining the oxidation state and content of the Cl_x^- anions, the observed peak shape change is sufficient to demonstrate the fluorinated electrolyte's antioxidant behavior under high voltage. Additionally, Zr 3d spectra exhibit minimal variation,

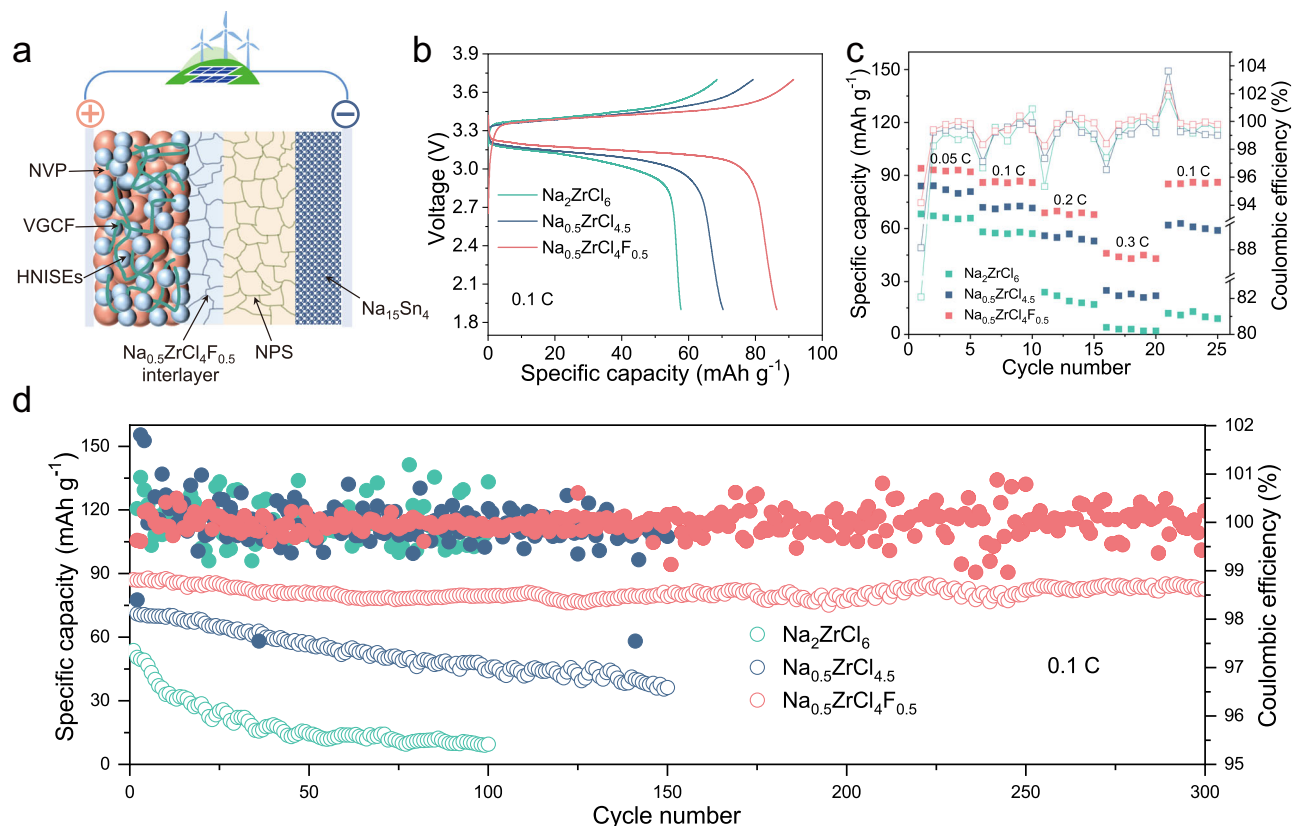


Fig. 5 | NVP-based all-solid-state batteries employing HNISEs. a Schematic illustration of HNISEs-based ASSIBs, comprising Na_2ZrCl_6 , $\text{Na}_{0.5}\text{ZrCl}_{4.5}$ and $\text{Na}_{0.5}\text{ZrCl}_4\text{F}_{0.5}$. **b** Charge-discharge curves at a rate of 0.1 C (11.7 mA g^{-1}). **c** Rate

performance of ASSIBs using various halide catholytes at RT, evaluated across different current densities. **d** Long-term cycling stability of ASSIBs tested at 0.1 C (11.7 mA g^{-1}).

indicating the maintenance of Zr in its highest oxidation state, Zr^{4+} (Supplementary Fig. 18).

In addition to impressive oxidation stability, $\text{Na}_{0.5}\text{ZrCl}_4\text{F}_{0.5}$ also demonstrates a decreased absolute reduction current of 0.0266 mA at 0 V (Supplementary Fig. 19), corroborating the improvement of intrinsic reduction stability against $\text{Na}_{15}\text{Sn}_4$ negative electrode by fluorination strategy. Further analysis of the interface kinetic stability between $\text{Na}_{0.5}\text{ZrCl}_4\text{F}_{0.5}$ and $\text{Na}_{15}\text{Sn}_4$ involved cycling $\text{Na}_{15}\text{Sn}_4 | \text{Na}_{15}\text{Sn}_4$ symmetrical cells. As illustrated in Supplementary Fig. 20, the initial plating/stripping potential of the $\text{Na}_{0.5}\text{ZrCl}_4\text{F}_{0.5}$ and $\text{Na}_{0.5}\text{ZrCl}_{4.5}$ -based cells were approximately 0.8 V at a current density of 0.1 mA cm^{-2} and a capacity of 0.1 mAh cm^{-2} . The $\text{Na}_{0.5}\text{ZrCl}_4\text{F}_{0.5}$ -based symmetrical cell maintained stable plating/stripping for 800 h with minimal overpotential increases (~ 1 V), whereas its $\text{Na}_{0.5}\text{ZrCl}_{4.5}$ -based counterpart delivered a progressive overpotential increased to approximately 2.6 V over the same duration. In contrast, the Na_2ZrCl_6 -based symmetrical cell displayed a appreciable overpotential increase and shorter plating/stripping times (exceeding 5 V within 73 h), primarily due to the lowest conductivity and poor reduction stability of the electrolytes. Additionally, the improved cycling stability in $\text{Na}_{15}\text{Sn}_4 | \text{Na}_{0.5}\text{ZrCl}_4\text{F}_{0.5} | \text{Na}_{15}\text{Sn}_4$ cells likely originates from the in-situ formation of a kinetically stable interphase layer between the fluorinated SEs and the negative electrode interface (e.g., NaF, see Supplementary Fig. 21), which helps mitigate ongoing parasitic reactions during cycling^{25,48–50}.

Although $\text{Na}_{0.5}\text{ZrCl}_4\text{F}_{0.5}$ -based symmetrical cells exhibit reduced plating/stripping overpotentials, these remain relatively high (~ 0.8 V) to effectively release the full battery's capacity. Therefore, the Na_3PS_4 (NPS, $-1.4 \times 10^{-4} \text{ S cm}^{-1}$ at 25 $^\circ\text{C}$), known for its commendable kinetic stability against Na-Sn alloys^{15,51}, was incorporated as an intermediate electrolyte layer between the HNISEs and the $\text{Na}_{15}\text{Sn}_4$ negative

electrode. Moreover, recent studies have shown that fluorination of halides is advantageous for promoting interfacial compatibility with sulfides²⁶. We explored this by comparing the electrochemical behaviors of two symmetrical cell configurations: both utilized HNISEs as interlayers and NPS as outer layers. As depicted in Supplementary Fig. 22, both configurations initially exhibited low overpotentials (~ 0.2 V). However, the $\text{Na}_{0.5}\text{ZrCl}_{4.5}$ -based sandwich symmetrical cell presented increasing plating/stripping potentials over time, signaling interfacial incompatibility between $\text{Na}_{0.5}\text{ZrCl}_{4.5}$ and NPS during cycling. In contrast, the $\text{Na}_{0.5}\text{ZrCl}_4\text{F}_{0.5}$ -based sandwich symmetrical cell maintained satisfactory cycling stability with negligible overpotential increase, underscoring the beneficial impact of fluorine doping on halide/sulfide interfacial compatibility. Based on these findings, a dual-electrolyte layer strategy was adopted for ASSIBs, positioning $\text{Na}_{0.5}\text{ZrCl}_4\text{F}_{0.5}$ adjacent to the positive electrode and NPS near the $\text{Na}_{15}\text{Sn}_4$ alloy negative electrode.

Electrochemical energy storage behaviors of ASSIBs

The electrochemical performance of mechanochemically synthesized $\text{Na}_{0.5}\text{ZrCl}_4\text{F}_{0.5}$, $\text{Na}_{0.5}\text{ZrCl}_{4.5}$, and Na_2ZrCl_6 HNISEs in ASSIBs was evaluated using $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ (NVP, with 117 mAh g^{-1} theoretical capacity) positive electrodes and VGCF conductive additives, as shown in Fig. 5a. Detailed material characteristics are provided in Supplementary Figs. 23–24. For comparative purposes, we constructed a single-layer cell configuration employing only NPS, without HNISEs (Supplementary Fig. 25a). As depicted in Fig. 5b, under a voltage window of 1.9 to 3.7 V at RT and a 0.1 C (11.7 mA g^{-1}) rate, the initial Coulombic efficiency (ICE) of the $\text{Na}_{0.5}\text{ZrCl}_4\text{F}_{0.5}$ cell was markedly higher at 94.5%, with a discharge capacity of 86.4 mAh g^{-1} , in contrast to the $\text{Na}_{0.5}\text{ZrCl}_{4.5}$ and Na_2ZrCl_6 cells, which achieved ICE of 88.7% and 84.2% with discharge capacities of 70.3 mAh g^{-1} and 57.7 mAh g^{-1} , respectively. The

enhanced electrochemical stability of the $\text{Na}_{0.5}\text{ZrCl}_4\text{F}_{0.5}$ electrolyte appears to shield the other HNISEs and NPS from oxidation through interaction with NVP. Nonetheless, the cell employing NPS electrolyte exhibited a rapid decline in discharge capacity and increased interfacial resistance following the initial charge, attributed to the poor cathodic stability under high voltages, as further detailed in Supplementary Figs. 25b–c⁵².

The rate and cycle capability of batteries employing $\text{Na}_{0.5}\text{ZrCl}_4\text{F}_{0.5}$, $\text{Na}_{0.5}\text{ZrCl}_{4.5}$, and Na_2ZrCl_6 as catholytes demonstrate comparable capabilities, as shown in Fig. 5c, d. Specifically, the $\text{Na}_{0.5}\text{ZrCl}_4\text{F}_{0.5}$ cell exhibits a higher specific capacity of 46.5 mAh g^{-1} at 0.3 C (35.1 mA g^{-1}), considerably outperforming the capacities of 25.3 and 4.6 mAh g^{-1} observed in the $\text{Na}_{0.5}\text{ZrCl}_{4.5}$ and Na_2ZrCl_6 cells, respectively. When the current density is reduced back to 0.1 C (11.7 mA g^{-1}), a reversible capacity of 82.3 mAh g^{-1} is restored. The improved rate capacity of the $\text{Na}_{0.5}\text{ZrCl}_4\text{F}_{0.5}$ cells primarily stems from their higher conductivity and enhanced compatibility with fluorinated electrolytes. It is noteworthy that the capacity limitations at higher rates are largely due to the suboptimal conductivities of the electrolyte layers, which are less than 1 mS cm^{-1} . The $\text{Na}_{0.5}\text{ZrCl}_4\text{F}_{0.5}$ -based cell also displays a substantial capacity retention of 94.4% over 300 cycles at 0.1 C (11.7 mA g^{-1}), markedly surpassing the performance of $\text{Na}_{0.5}\text{ZrCl}_{4.5}$ and Na_2ZrCl_6 cells, which retain only 49.5% and 17.7% of their initial capacities after 150 and 100 cycles, respectively. The fluctuations in cycling performance are mainly caused by the battery performance tests conducted at ambient temperatures, where capacity can be slightly influenced by temperature variations due to weather and seasonal changes. Furthermore, the lower overpotential and enhanced cycling reversibility of the $\text{Na}_{0.5}\text{ZrCl}_4\text{F}_{0.5}$ cell are further demonstrated in the charge/discharge voltage profiles and corresponding dQ/dV integral curves (Supplementary Figs. 26–27).

The EIS tests were conducted to assess the interfacial stabilities between positive electrodes and HNISEs in both pre- and post-long-term cycling. As illustrated in Supplementary Fig. 28, Nyquist plots present two distinct semicircles at high and mid frequencies, corresponding to the impedance of separating electrolyte layers and composite positive electrode interfaces, respectively⁵². Following extended cycling, the $\text{Na}_{0.5}\text{ZrCl}_4\text{F}_{0.5}$ cell ($\sim 711 \Omega$) demonstrated lower interfacial resistance compared to $\text{Na}_{0.5}\text{ZrCl}_{4.5}$ ($\sim 4237 \Omega$) and Na_2ZrCl_6 ($\sim 7346 \Omega$) cells, indicating better electrochemical stability of $\text{Na}_{0.5}\text{ZrCl}_4\text{F}_{0.5}$ catholytes during cycling. Moreover, the in-situ galvanostatic-EIS (GEIS) combined with the distribution of relaxation time (DRT) method was performed to evaluate the detailed electrochemical evolution of HNISEs cells during charging. Supplementary Fig. 29a–c show no distinct changes in resistance during the charging process (2.9 to 3.7 V) for the $\text{Na}_{0.5}\text{ZrCl}_4\text{F}_{0.5}$ cell, yet notable impedance growth is observed in $\text{Na}_{0.5}\text{ZrCl}_{4.5}$ and Na_2ZrCl_6 analogs. Semi-quantitative DRT results (Supplementary Fig. 29d–f) categorize relaxation time (τ) into four groups—S1, S2, S3, and S4—across these three cells. Peak changes in the S3 ($10^{-4} < \tau < 10^0 \text{ s}$) and S4 ($10^0 < \tau < 10^1 \text{ s}$) regions are predominantly linked to the electrolyte/positive electrode interface and diffusion resistance, respectively⁵³. For the $\text{Na}_{0.5}\text{ZrCl}_4\text{F}_{0.5}$ cell, peak intensities in the S3 and S4 regions show minimal changes with relaxation time, underscoring good electrochemical stability within composite positive electrodes. Conversely, peak intensities in S3 and S4 regions increase notably with relaxation time for $\text{Na}_{0.5}\text{ZrCl}_{4.5}$ and Na_2ZrCl_6 cells, indicative of irreversible parasitic reactions at electrolyte/positive electrode interfaces, leading to deteriorating interface stability and heightened cell impedance. In conclusion, based on EIS and DRT analyses, $\text{Na}_{0.5}\text{ZrCl}_4\text{F}_{0.5}$ catholytes exhibit preferable high-voltage stability and interface compatibility with polyanionic-type positive electrodes, thereby boosting ASSIBs' performance.

To investigate the influences of chemical-mechanical failure on full cells, we compared the micro-morphological evolution of composite positive electrode surfaces using SEM images of pristine and

cycled samples. Figure 6a illustrates that all three electrode types show similar microcrack areas on their surfaces before cycling, manifesting insufficient connection between HNISEs and positive electrode materials. After several hundred cycles, the $\text{Na}_{0.5}\text{ZrCl}_4\text{F}_{0.5}$ -based composite electrodes delivered a uniform and dense structure with no discernible microcracks, in contrast to the other two types, which displayed more and longer cracks. This conversion stems from the continuous enhancement of $\text{Na}_{0.5}\text{ZrCl}_4\text{F}_{0.5}|\text{NVP}$ interfacial mechano-chemical stability during cycling, alongside the inherent high deformability of fluorinated SEs. Additionally, Na-ion diffusion coefficients (D_{Na^+}) for three configurations were calculated through the galvanostatic intermittent titration technique (GITT), a standard method for assessing ion transport kinetics within composite cathode⁵⁴. As demonstrated in Fig. 6b and Supplementary Fig. 30, the $\text{Na}_{0.5}\text{ZrCl}_4\text{F}_{0.5}$ -based positive electrode composites display higher D_{Na^+} and lower overpotential during charging, affirming the favorable kinetic and thermodynamic stability of $\text{Na}_{0.5}\text{ZrCl}_4\text{F}_{0.5}$ catholyte within the electrodes. Furthermore, the calculated interfacial reaction energies between HNISEs and electrodes provide further support for these observations (Fig. 6c and Supplementary Table 10). The $\text{Na}_{0.5}\text{ZrCl}_4\text{F}_{0.5}$ ($\sim 84 \text{ meV}$) catholytes and their phase equilibria products (Na_2ZrF_6 $\sim 15.5 \text{ meV}$) exhibit a lower interface reaction driving force with NVP cathodes compared to the $\text{Na}_{0.5}\text{ZrCl}_{4.5}$ ($\sim 88 \text{ meV}$) counterpart, suggesting a more favorable interfacial compatibility between the fluorinated electrolyte and NVP. It is noteworthy that the mutual reaction energies between HNISEs and negative electrodes (Na metal and Na_5Sn_4) are significantly higher than those at the positive electrode interface, highlighting the inherent thermodynamic instability of the negative electrode|HNISE interface. However, favorable interfacial decomposed insulating phases (such as NaF and NaCl) can be formed from fluorinated SEs during cycling, which helps inhibit the interfacial parasitic reactions and the continuous decomposition of HNISEs^{25,55}. Thus, fluorinated SEs demonstrate greater stability at the negative electrode interface, consistent with the stable cycling observed in $\text{Na}_{0.5}\text{ZrCl}_4\text{F}_{0.5}$ -based symmetrical cells (Supplementary Fig. 20). In a word, the Na halide SEs could achieve improved RT conductivities using the Na/Cl-DC strategy owing to an effective balance of vacancy and carrier concentrations within their crystal structures. Moreover, the introduction of fluorine into structures further enhanced their (electro)chemical stability and interface compatibility. Surprisingly, the fluorinated electrolyte also demonstrated boosted conductivity, deformability, and moisture stability, largely attributed to the amorphous effect and stronger bonding within structures (as summarized and illustrated in Fig. 6d and Supplementary Fig. 31).

In conclusion, we have proposed a vacancy and carrier concentrations-modulated design strategy for enhancing ion conduction in Na halide-based SEs. This approach optimally adjusts vacancy and carrier concentrations within the lattice, establishing critical balancing concentration thresholds that enhance Na^+ diffusion efficiency. As a result, the conductivities of these Na/Cl-DC SEs exceed those of their standard counterparts. Furthermore, integrating fluoride into zirconium-based chloride SEs not only augments their electrochemical stability and interface compatibility with electrodes but also increases conductivity and moisture tolerance. This improvement is primarily attributed to the formation of an amorphous phase and the strengthened Zr-F bonds within the electrolyte matrix. The fluorination-driven amorphization boosts local chemical structural disorder and increases the prismatic Na coordination environment in the structure, thus improving Na^+ mobility. Employing NVP as a positive electrode and $\text{Na}_{0.5}\text{ZrCl}_4\text{F}_{0.5}$ as a catholyte, ASSIBs exhibit good cycling stability, maintaining approximately 94% of their initial capacity after 300 cycles at a discharge rate of 0.1 C (11.7 mA g^{-1}). This study provides an alternative proposal for the design of advanced Na-ion halide SEs, optimizing both conductivity and (electro)chemical stability, thus promoting the durability and safety of ASSIBs.

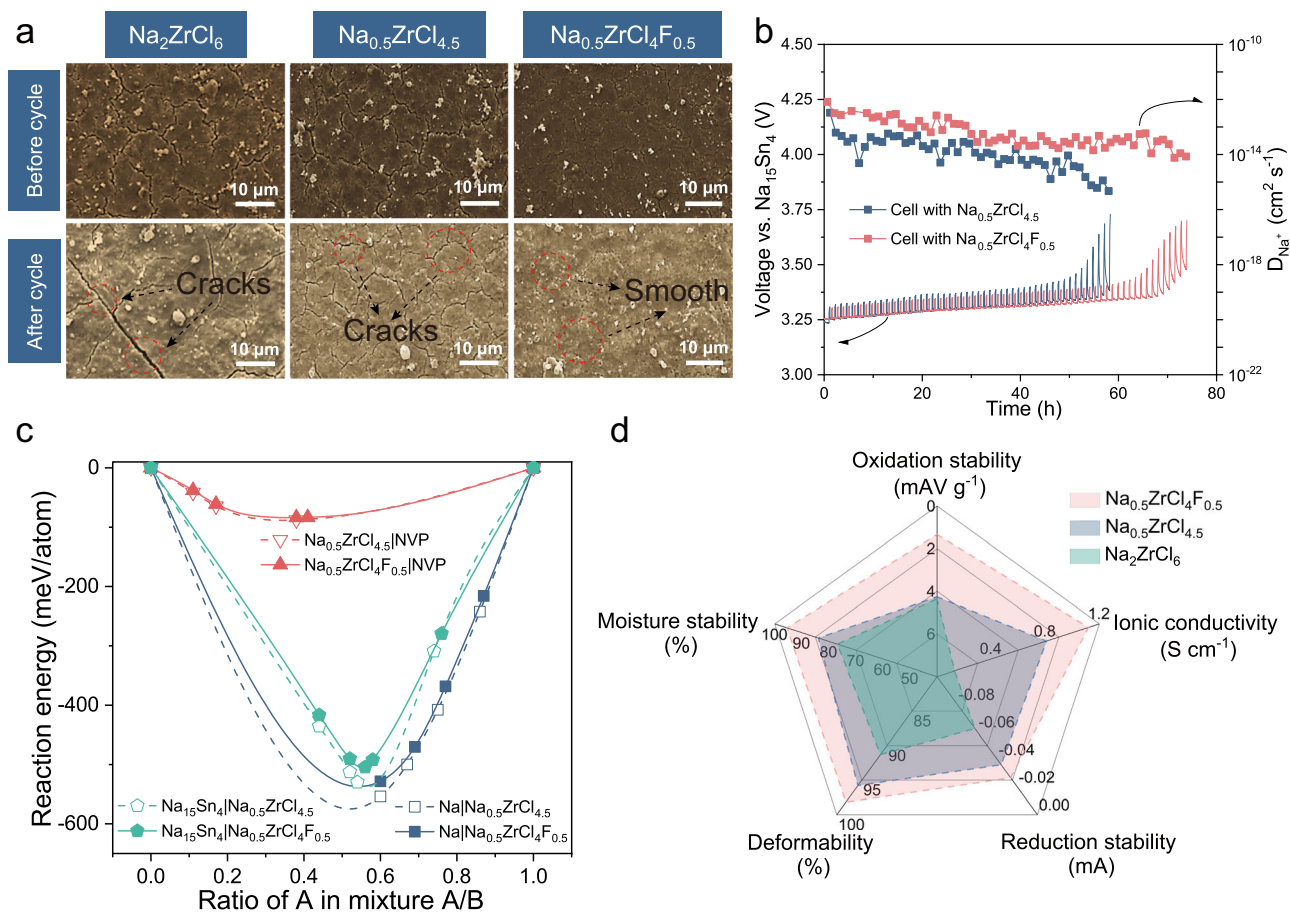


Fig. 6 | Analysis and summary of HNISEs. **a** Surface micro-morphological evolutions in positive electrode composites before and after cycling. False coloring was applied to the images to clearly emphasize the changes in the micro-morphology of the samples. **b** GITT curves and comparative analysis of ion diffusion coefficients in

HNISEs-based ASSIBs. **c** Calculated mutual reaction energies between HNISEs and electrode interfaces. **d** Spider plot comparing the properties of Na_2ZrCl_6 , $\text{Na}_{0.5}\text{ZrCl}_{4.5}$, and $\text{Na}_{0.5}\text{ZrCl}_4\text{F}_{0.5}$ SEs. The moisture stability of HNISEs was quantified by comparing the retention of conductivities before and after air exposure.

Methods

Materials synthesis

High-purity anhydrous powders, specifically NaCl (Aladdin, 99.99%), ZrCl_4 , ZrF_4 , YbCl_3 , and ErCl_3 (each from Aladdin at 99.9%), served as the precursor materials. The series sodium chloride-based SEs, designated as $\text{Na}_x\text{ZrCl}_{4.5+x}$, $\text{Na}_x\text{Er}_{0.4}\text{Zr}_{0.6}\text{Cl}_{3.6+x}$, $\text{Na}_x\text{Yb}_{0.25}\text{Zr}_{0.75}\text{Cl}_{3.75+x}$, and the fluorinated variant $\text{Na}_{0.5}\text{ZrCl}_{4.5-y}\text{F}_y$, were synthesized through high-energy ball-milling under an Ar atmosphere. This process employed a 20:1 zirconia ball-to-precursor mass ratio at 600 rpm over 10 h in a zirconia ball mill jar, adhering to stoichiometric guidelines. Similarly, Na_3PS_4 (NPS) SEs were produced by ball-milling a stoichiometric mixture of Na_2S (Aladdin, 95%) and P_2S_5 (Macklin, 99%) under the same conditions, followed by annealing at 270 °C for 2 h within vacuum quartz tubes (vacuum level: 5×10^{-5} Pa). The NVP positive electrode materials were synthesized via a solid-state reaction following established methods³⁶. A stoichiometric mixture of NaH_2PO_4 and V_2O_3 (Aladdin, >99%) was ball-milled at 400 rpm for 10 h and then dried at 60 °C for 24 h in an oven. Subsequently, the powders were sintered at 900 °C for 24 h in an argon atmosphere.

Characterizations

X-ray diffraction (XRD) analysis was conducted using a D/MAX-RB Rigaku device with Cu K α radiation to identify material phases and structures. To minimize air exposure, Kapton tapes were used to cover the sample surfaces. The chemical composition of prepared samples was characterized by an inductively coupled plasma optical emission spectrometer (ICP-OES, Agilent 5110). Phase quantification was refined

using the Rietveld method with GSAS II software⁵⁷. The morphology and elemental composition of the samples were examined via scanning electron microscopy (SEM, JEOL JSM-7001F) and energy dispersive spectroscopy (EDS, Oxford). Porosity and relative density estimates were derived from SEM images, using the Adjust-Threshold plugin in ImageJ⁵⁸. For high-resolution transmission electron microscopy (HRTEM) analysis, the samples were placed on a lacey Cu grid, and images were captured using a JEM-2100 microscope. Raman spectroscopic data were obtained with a JY HR800 confocal Raman microscope. Chemical compositions and states were assessed through X-ray photoelectron spectroscopy (XPS, ESCALAB 250Xi, Thermo Fisher). ^{23}Na solid-state nuclear magnetic resonance (ss-NMR) measurements were performed using a Bruker Avance Neo 400WB spectrometer (9.40-T wide-bore magnet, Larmor frequency of 105.84 MHz for ^{23}Na) equipped with a 4-mm MAS probe at room temperature. The resonance frequency for ^{23}Na was set at 100.62 MHz, with a rotation frequency of 10 kHz. The chemical shift reference was calibrated to 0 ppm using NaCl powder spun at 10 kHz. To prevent exposure to moisture in the ambient atmosphere, all samples were prepared within an argon-filled glovebox. Zr K-edge synchrotron X-ray Absorption Fine Structure (XAFS) measurements were performed at the BL13SSW beamline of the Shanghai Synchrotron Radiation Facility, with spectra acquired in fluorescence mode. The resulting data were analyzed using the Demeter software package⁵⁹.

Ionic conductivities of synthesized SEs at temperatures ranging from 25 to 80 °C were measured using alternating current (AC) impedance spectroscopy using two stainless steel rods as blocking

electrodes, over a frequency range of 0.1 Hz to 1 MHz, at a 10 mV amplitude, utilizing a Solartron 1260 + 1287 electrochemical station. The SEs powders (~150 mg) were cold-pressed into pellets with a diameter of 10 mm at 300 MPa. Electronic conductivities were determined via the direct current (DC) polarization technique at 25 °C, applying a 1 V bias for 1800 s with stainless steel ion-blocking electrodes. Electrochemical stability was assessed through cyclic voltammetry (CV) using a CHI660B workstation. Measurements were conducted in a Na₁₅Sn₄|NPS|HNISEs|HNISEs-VGCF (vapor-grown carbon fibers) cell configuration within a 0 to 5 V range (vs. Na|Na⁺) at a scan rate of 0.1 mV s⁻¹. The mass ratio of SEs to VGCF was maintained at 7:3, with an approximate hybrid mass loading of 12.7 mg cm⁻². A symmetric cell configuration of Na₁₅Sn₄|SEs|Na₁₅Sn₄ was constructed to evaluate interfacial stability against the Na₁₅Sn₄ negative electrode, testing at a current density of 0.1 mA cm⁻² and a capacity of 0.1 mAh cm⁻². Interfacial and full-cell resistances across different catholytes were characterized by electrochemical impedance spectroscopy (EIS). The model ASSIBs were assembled using Na₃V₂(PO₄)₃ (NVP) as the positive electrode material, Na₁₅Sn₄ as the negative electrode, and a combination of halides SEs and/or NPS as the electrolyte separator layers. The positive electrode composites were prepared by ball-milling NVP, SEs, and VGCF in a mass ratio of 45:50:5 at 300 rpm for 30 min. The Na₁₅Sn₄ alloy negative electrode was prepared through the ball-milling of sodium metal (Aladdin, 99.9%) and tin powder (Aladdin, 99.99%) at 300 rpm for 20 h under an Ar atmosphere. Dense electrolyte layers, 10 mm in diameter, were formed by compressing 40 mg of Na_{0.5}ZrCl₄F_{0.5} and 60 mg of NPS at 100 MPa. Subsequently, 10 mg of composite positive electrode material and 30 mg of Na₁₅Sn₄ powder were uniformly spread on either side of the electrolyte pellets and compacted at 370 MPa for 2 mins. All cells are assembled in an Ar-filled glove box (O₂ < 0.1 ppm, H₂O < 0.1 ppm). Galvanostatic charge-discharge cycles for the ASSIBs were conducted between 1.9 and 3.7 V versus Na-Sn|Na⁺ at RT and in an air environment using a LAND CT2001A cell test system under a constant pressure of 70 MPa. Galvanostatic intermittent titration technique (GITT) tests were carried out at a rate of 0.1 C (11.7 mA g⁻¹), with each charging step lasting 10 min followed by a one-hour rest period. EIS assessed both full cell and interfacial resistances across various catholytes, employing a frequency range from 0.1 Hz to 10⁶ Hz at a 10 mV amplitude. In-situ EIS measurements were performed at various potentials (2.9, 3.1, 3.3, 3.5, and 3.7 V versus Na-Sn|Na⁺) at a rate of 0.1 C (11.7 mA g⁻¹). To ensure the reproducibility of results, a minimum of three cells were employed in each electrochemical experiment.

Theoretical calculation

Bond valence site energy (BVSE) calculations were carried out using the softBV⁶⁰ program, which utilized structural models derived from XRD Rietveld refinement. The program assessed the energies of various Na sites within the crystal structures on a 3D grid with 0.1 Å resolution, employing the transferable Morse-type softBV force field. Zones of low BVSE, indicative of potential Na-ion conduction pathways, were identified. VESTA⁶¹ software was used to visualize the crystal structures and the potential energy isosurfaces.

The interface pseudo-binary reaction energies were calculated as:

$$\Delta E_D(\text{SSE, electrodes}, x) = E_{\text{eq}}(C_{\text{interface}}(C_{\text{SEs}}, C_{\text{electrodes}}, x)) - E_{\text{interface}}(\text{SSE, electrodes}, x)$$

Where C_{SEs} and $C_{\text{electrodes}}$ represent the compositions of SEs and electrode materials, respectively, normalized to one atom per formula unit. The variable (x) denotes the molar fraction of the SEs. Material energies were obtained from the Materials Project database⁶².

Density functional theory (DFT) calculations were performed using the Vienna Ab initio Simulation Package⁶³ via employing the

generalized gradient approximation parameterized by Perdew-Burke-Ernzerhof (PBE) exchange-correlation function⁶⁴. To model the Na_{0.5}ZrCl₄F_{0.5} amorphous structure, ab initio molecular dynamics (AIMD) simulations were conducted at 300 K, followed by annealing cycles from 300 K to 1800 K and back to 300 K. A plane-wave energy cut-off of 300 eV was applied to balance computational efficiency. The time step for the AIMD simulations was set to 1 fs, with a total of 4000 steps corresponding to 4 ps of simulation time in an NPT ensemble (constant particle number, pressure, and temperature). An annealing rate of 0.5 K/fs was used.

Data availability

The data generated and analyzed in this study are provided within the paper and its Supplementary Information. All additional information is available from the corresponding authors. Source data are provided with this paper.

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Author contributions

L.Z.F. and C.W.N. conceived the project. M.W. designed and conducted the experiments, including materials synthesis, characterization, and data analysis. X.L. and L.G. contributed to data interpretation via computational studies. J.Z. performed the synchrotron X-ray characterization. H.L., D.L., and X.Q. provided support in electrochemical testing and manuscript revision. All authors engaged actively in discussion and iterative refinement of the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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